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Pixel-wise skin colour detection based on flexible neural tree

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Abstract: Skin colour detection plays an important role in image processing and computer vision. Selection of a suitable colour space is one key issue. The question that which colour space is most appropriate for pixel-wise skin colour detection is not yet concluded. In this study, a pixel-wise skin colour detection method is proposed based on the flexible neural tree (FNT) without considering the problem of selecting a suitable colour space. A FNT-based skin model is constructed by using large skin data sets which identifies the important components of colour spaces automatically. Experimental results show improved accuracy and false positive rates (FPRs). The structure and parameters of FNT are optimised via genetic programming and particle swarm optimisation algorithms, respectively. In the experiments, nine FNT skin models are constructed and evaluated on features extracted from RGB, YCbCr, HSV and CIE-Lab colour spaces. The Compaq and ECU datasets are used for constructing FNT-based skin model and evaluating its performance compared with other skin detection methods. Without extra processing steps, the authors method achieves state of the art performance in skin pixel classification and better performance in terms of accuracy and FPRs.

1 Introduction

Skin colour detection plays an important role in image processing and video analysis, especially in human–computer interaction techniques. Skin colour detection provides initial search spaces for objects of interest, such as faces in face detection and face localisation, or hands in hand tracking and gesture recognition.

For the attractive properties of high processing speed, invariance against rotation, partial occlusion and pose change, skin colour is a popular and powerful cue in human detection and tracking. In recent years, a number of pixel-based skin colour detection methods are published [1]. Although skin colour would be affected by changing lighting conditions, complex backgrounds or objects having skin-like colours, numerous methods adopt skin detection as a preliminary step in computer vision applications. For example, Han *et al.* [2] built a skin colour model to perform segmentation and tracking of skin objects from videos in sign language recognition. Jin *et al.* [3] detected faces in colour images via combination of extracting skin-region and template matching method. Skin colour detection could provide candidates for other algorithms. Vizireanu [4] proposed the method of generalisation of binary morphological shape decomposition, and skin colour detection would provide the binary image (or grey image) for morphological shape decomposition representation or the inter-frame interpolation method [5]. Other applications like hand gesture recognition [6] and naked image detection [7], also exploit skin colour detection.

The objective of pixel-wise skin colour detection is to build a decision rule that classifies each pixel as skin or non-skin individually. There are two primary steps for skin colour detection: the choice of colour spaces and classification techniques. Skin detection methods represent image pixels in a suitable colour space and build skin colour models according to the distributions of components in different colour spaces. People have given comparative studies and investigation of the effect of colour space choice on skin detection performance [8, 9], which shows that the choice of colour space not only affect the performance of skin detection, but also determines how effectively we can model the skin colour distribution. Skin classification techniques is another key issue which can be roughly divided into three steams: fixed threshold methods, parametric methods and non-parametric methods. People have surveyed the pixel-wise skin classification methods and presented comparative studies [1, 10]. Algorithms in machine learning, such as support vector machine (SVM), multilayer perceptron (MLP) and Adaboost have also been investigated for skin colour detection [11].

Most of skin detection methods have gained success in practice. However, there are problems with the state of the art methods: (i) the question that which colour space is most suitable for skin colour detection is not concluded yet. The performance of skin colour classifiers is significantly

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affected by the choice of colour spaces; (ii) methods with higher accuracy always have higher false positive rate (FPR), which makes difficulty in the detailed detection and high level processing. To address these problems, a flexible neural tree (FNT) [12, 13] based skin colour model is proposed based on our previous work [14, 15]. The proposed method has two contributions: (i) automatic selection of important components of colour spaces. The choice of colour spaces is not a problem here, because FNT based skin colour model allows selection of input variables. Given all components of colour spaces, the proposed method enable automatic selection of the most important colour channels from various colour representations; (ii) better performance than state of the art methods in terms of accuracy and FPR. Individuals with simple structure and similar accuracy models will be selected and reserved in the training stage, which makes our method giving higher accuracy and lower FPR than other skin detection methods.

The rest of the paper is organised as follows. In Section 2, we discuss related work. The FNT and the novel skin colour model is presented in Section 3. In Section 4, the experimental results are shown with analysis, and conclusions are given in Section 5.

2 Related work

Most of researches on skin detection focus on two primary problems: the selection of suitable colour spaces and the classification techniques.

2.1 Selection of colour spaces

Colour space is a method by which we can specify, create and visualise colour. A general colour is defined by its attributes of brightness, hue and colourfulness. Thus, a colour is usually specified using three co-ordinates or parameters. These parameters describe the position of the colour within the colour space [16]. For skin colour detection, numerous colour spaces are used to represent pixel distributions, and many of them share similar characteristics. The following lists the commonly used colour spaces in skin colour detection.

- *RGB (Red Green Blue).* RGB is very common, being used in skin detection [17–21], since image data captured by camera are normally provided and stored as RGB. For example, Yang *et al.* [17] constructed two codebooks using colour (RGB) features and local binary pattern features for visual tracking.
- *HSL (hue saturation and lightness).* This represents a wealth of similar colour spaces, alternative names include HSI (intensity), HSV (value) etc. Most of these colour spaces are linear transforms from RGB [22, 23].
- *YIQ, YUV, YCbCr and YCC (luminance–chrominance).* These colour spaces separate RGB into luminance and chrominance information. The YCbCr space is one of the most popular and commonly used choice for skin detection [7, 24, 25].
- *CIE.* There are three CIE-based colour spaces in skin detection, that is, CIE-XYZ [26], CIE-Luv [27] and CIE-Lab [28]. They are often used for evaluation and comparison with other colour spaces [9].
- *Others colour spaces.* To reduce the dependence on lighting and obtain reduction in dimensionality, the RGB components are normalised to generate normalised RGB (*rgb*) [29] colour space. IHLS [30] and new colour spaces [31] also appeared in published work.

It is a challenging task to determine suitable colour space for skin detection. Literatures giving comparative studies and investigations of the effect of colour space choice on skin detection performance are reported in recent years. Phung *et al.* [9] investigate eight different colour representations using the Bayesian classifier with the histogram technique. They reported that feature vectors comprising all channels outperformed feature vectors containing only chrominance channels. In order to know which colour space is the best option for building an efficient face detector, Chaves *et al.* [8] studied ten colour spaces by doing comparisons among them. In concrete, they studied seven models and said that HSV model was the winner in their study. In the work of Khan *et al.* [11], six colour spaces (IHLS, HSI, RGB, normalised RGB, YCbCr and CIE-Lab) with the presence or absence of the illuminance component were compared and evaluated using machine learning algorithms. Furthermore, many works dropped the luminance components of colour spaces, used only chrominance and expected their methods more robust to lighting variations. However, the published work have shown that the absence of illuminance components decreased the skin detection performance [10, 11].

To the best of our knowledge, there is no agreement on the question that which colour space is the best option for human skin detection. The selection of colour spaces is still a challenging problem for skin detection methods.

2.2 Classification techniques

The goal of skin pixel classification techniques is to build decision rules to discriminate between skin and non-skin pixels. There are numerous skin classification methods, which have been surveyed in published work [1, 10, 30]. These methods can be roughly divided into the following categories: explicitly defined skin region, non-parametric skin distribution, parametric skin distribution and dynamic skin models.

- Explicitly defined skin region (fixed threshold) methods attempt to define explicitly boundaries of skin clustering in specific colour spaces through a number rules [10]. For example, rules: $77 \leq \mathrm{Cb} \leq 127$ and $133 \leq \mathrm{Cr} \leq 173$ define a fixed skin range in CbCr space [9]. Methods in this category have rapid processing speed, but have difficulty in finding both good colour spaces and adequate decision rules to achieve high accuracy.
- Non-parametric methods estimate skin colour distribution from training data without generating an explicit skin model. The colour space is quantised into a number of bins that store the number of times this particular colour occurred. In Bayesian classifier [32], $P(C|\mathrm{skin})$ are the respective class-conditional probability density function (pdf) of skin and non-skin colours, which can be directly computed from skin and non-skin colour histograms. Given colour value c, $P(\mathrm{skin}|C)$ will be computed via Bayes rule. Then, $p(\mathrm{skin}|c) \geq \Theta$ can be used as a skin detection rule using a threshold value Θ. Although non-parametric methods are fast in training and usage and theoretically independent to the shape of skin distribution, they require much storage to maintain skin probability map and their performance directly depends on the training data.

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• Mixture of Gaussians [21, 33] and elliptic boundary model [34] belong to parametric method. They model skin colour distribution via joint pdf like elliptical Gaussian, which makes these methods having ability of better generalisation from limited training data than other methods. Mixture of Gaussians has a form of pdf as $\sum_{i=1}^{k} \pi_i\, p_i(c|\text{skin})$, where k is the number of mixture components, π_i is the mixture parameter with constraint $\sum_{i=1}^{k} \pi_i = 1$, and $p(c|\text{skin})$ is Gaussian pdf. The elliptic boundary model is fast and simple with the considering that skin colour cluster is approximately elliptic in shape.

Most of methods mentioned above have achieved good performance in terms of accuracy (Acc), true positive rate (TPR), FPR and F-score. According to the reported results, histogram-based Bayesian classifier has 80% TP with 8.5% FP in original paper [32], and 93.4% TP with 19.8% FP in other paper [35] with RGB colour space. Elliptical boundary model obtained 90% TP with 20.9% FP in YCbCr colour space [34]. Gaussian mixture models in RGB colour space achieved 90% TP with 15.5% FP, whereas 90% TP with 30% FP in YIQ colour space [32]. Machine learning algorithms, such as MLP, have also been investigated [11]. Although state of the art skin detection methods give high TPR, it also produce high FPR. This is a crucial problem in details' detection, which will generate a poor candidate region for face detection or hand tracking.

3 FNT-based skin colour model

The FNT showed good performance in prediction [12, 13, 36]. Based on FNT and our previous work [14, 15, 37], a FNT-based skin colour model is proposed to address the problem of selection of colour spaces and improving performance in skin colour detection.

3.1 Flexible neural tree

FNT is a special multi-layer feed forward network [12]. A FNT can be generated by function set F and terminal instruction set T, which is described as follows

$$S = F \cup T = \{+_2,\ +_3,\ \ldots,\ +_N\} \cup \{x_1,\ \ldots,\ x_n\} \quad (1)$$

Function set F contains multi-category non-leaf nodes named flexible neurons (Fig. 1a). $+_i (i = 1, 2, 3, \ldots, N)$ means instructions of non-leaf nodes taking i inputs. The terminal instruction set T with elements $x_1, x_2, \ldots, x_n$ are instructions of leaf nodes taking no other arguments. A general flexible neuron operator is illustrated in Fig. 1a. Fig. 1b shows a typical instance of flexible neuron: the neuron operator has a form of '$+_3$' with three inputs and three weights 'w_1, w_2, w_3' representing the connection strength between a node

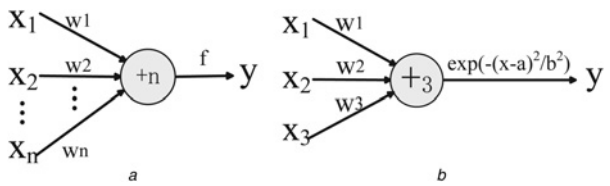

Fig. 1 *Flexible neuron*

a Flexible neuron operator
b Instance of flexible neuron

Table 1 Activation functions

Activation function	Formula
Gaussian function	$f(x, a, b) = (-(((x - a)^2)/(b^2)))$
unipolar sigmoid function	$f(x, a) = ((2/\lvert a\rvert)/(1 + e^{-2/\lvert a\rvert x}))$
bipolar sigmoid function	$f(x, a) = ((1 - e^{-2xa})/(a(1 + e^{-2xa})))$
non-local radial coordinates	$F(x, a, b) = (b^2 + \lVert x - a\rVert)^{-\alpha}\ (\alpha > 0)$

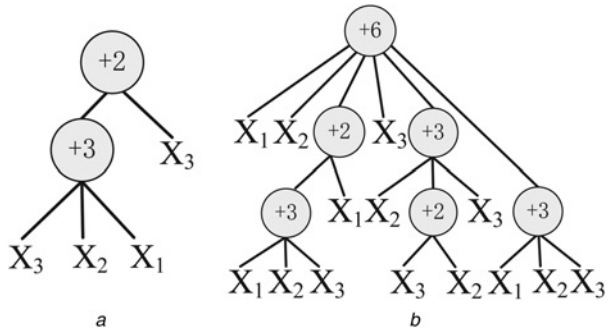

Fig. 2 *Structure of FNT models*

a '$+_2$' node
b '$+_6$' node

and its children. Activation functions listed in Table 1 can be used to compute the output of this neuron.

Given predefined model structure M_i, which has function set F_i and terminal instruction set Ti, the hierarchical structure and parameters of FNT model can be evolved and optimised using genetic programming (GP) algorithm [38] and particle swarm optimisation (PSO) algorithm [39], respectively. Fig. 2a shows a trained FNT model with function set $F = \{+_2, +_3\}$ and terminal instruction set $T = \{x_1, x_2, x_3\}$ (same as $M1$ model in Table 3). Another trained FNT model with $F = \{+_2, +_3, \cdots +_6\}$ and $T = \{x_1, x_2, x_3\}$ is illustrated in Fig. 2b.

In the stage of FNT modelling, random structures and corresponding parameters are evolved and optimised recursively, until a satisfactory solution is found or a time limit is reached. In addition, adjustable parameters, that is, a_i and b_i in Gaussian activation function, are randomly created as activation function parameters. Various activation functions can be used for FNT optimisation. The commonly used activation functions are listed in Table 1.

The overall output of FNT model is the total excitation of the root flexible neuron (i.e. '$+_2$' node in Fig. 2a, and '$+_6$' node in Fig. 2b. If Gaussian activation function is used, the excitation of '$+_n$' is $\text{net}_n = \sum_{j=1}^{n} w_j \cdot x_j$, where x_j ($j = 1, 2, \ldots, n$) are the input features to node '$+n$'. Combined with Gaussian function, the output of the node '$+_n$' is calculated by

$$\text{out}_n = f(a_n,\ b_n,\ \text{net}_n) = e^{-((\text{net}_n - a_n)/b_n)^2} \quad (2)$$

Thus, the overall output of FNT model can be computed from left to right by depth-first method recursively.

3.2 FNT-based skin colour modelling

There are three important issues for FNT-based skin colour modeling: optimisation of structure, optimisation of parameters and general learning procedure, which are depicted in Algorithm 1 (see Fig. 3) through Algorithm 3.

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Algorithm 1:

```
Data: M=Population Size; T=Maximum of Run
Result: Optimal Structure
initialise population P(0) randomly;
t = 0;
while t <= T do
    for i=1 to M do
        Evaluate fitness of P(t);
    end
    for i=1 to M do
        Select operation to P(t);
    end
    for i=1 to M/2 do
        Crossover operation to P(t);
    end
    for i=1 to M do
        Neural tree mutation operation to P(t);
    end
    for i=1 to M do
        P(t+1) = P(t);
    end
    t = t + 1;
end
```

Fig. 3 *Structure optimisation with GP*

In initialisation, feature x_i comprised of all components or channels c_i of commonly used colour spaces, written as $x_i = \{c_1, c_2, ..., c_{n_d}\}$, where n_d is the number of all channels of colour spaces, that is, the dimension of feature x_i. To find the optimal model, the normalised-mean-squared error (NMSE)

$$\mathrm{NMSE} = \frac{1/N_s \sum_{i=1}^{N_s} (x_i - \hat{x}_i)^2}{1/N_s \sum_{i=1}^{N_s} (x_i - \bar{x})^2} \tag{3}$$

or root-mean-squared error (RMSE)

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N_s} (x_i - \hat{x}_i)^2}{N_s}} \tag{4}$$

will be employed as fitness function, where the N_s is the number of samples, x_i denotes the actual skin data, $\hat{x}_i$ is the output of the skin model and $\bar{x}$ is the average of skin data.

3.2.1 Structure optimisation with GP: The optimisation of FNT-based skin model includes structure optimisation and parameter optimisation. Besides the standard operators of crossover and selection [38], special neural tree mutation operators are employed to generate offspring from the parents in the FNT-based skin modeling [12]. Some mutation operators are listed as follows:

1. *Changing one terminal node:* randomly select one terminal node in the neural tree and replace it with another terminal node.
2. *Changing all the terminal nodes:* select each terminal node in the neural tree and replace it with another terminal node.
3. *Growing:* select a random leaf in hidden layer of the neural tree and replace it with a newly generated subtree.
4. *Pruning:* randomly select a function node in the neural tree and replace it with a terminal node.

As listed in Algorithm 1 (see Fig. 3), structure optimisation using GP starts with an initial population composed of F and T sets. A set of randomly initialised FNTs generates the population $P(0)$. The fitness function provided in (3) and (4), will be used to measure or compare how well it performs the task of each individual program. Genetic operations, such as crossover, mutation and reproduction, are applied to individual(s), that is, neural trees of population $P(0)$. The individual will be probabilistically selected from the population based on fitness. In this probabilistic selection process, better individuals are favoured over inferior individuals. After the genetic operations are performed on the current population, the population of offspring (i.e. the new generation) replaces the current population (i.e. the now–old generation). This iterative process of measuring fitness and performing the genetic operations is repeated over many generations (t denotes number of generation in Algorithm 1 (see Fig. 3). The run of genetic programming terminates when the termination criterion is satisfied. The outcome of the run is specified by the method of result designation. The best individual ever encountered during the run (i.e. the best-so-far neural tree) is typically designated as the result of the run.

3.2.2 Parameter optimisation with PSO: When an optimal or near-optimal structure of FNT-based skin model is found via Algorithm 1 (see Fig. 3), its parameters, that is, the weights $\{w_j\}$ and flexible activation function parameters, will be optimized using PSO algorithm [13, 39].

PSO algorithm is a population-based stochastic optimisation technique, which is used to solve the optimisation problems. In PSO, the potential solutions, called particles, fly through the problem space by following the current optimum particles. All of particles have fitness values which are evaluated by the fitness function f to be optimised, and have velocities which direct the flying of the particles. PSO is initialised with a group of random particles (solutions) and then searches for optima by updating generations. In every iteration, each particle is updated by two best values. The first one is the best solution (fitness) it has achieved so far. This value is called $\boldsymbol{pb}$. Another best value that is tracked by the particle swarm optimiser is the best value obtained so far by any particle in the population. This best value is a global best and called $\boldsymbol{gb}$ After finding the two best values, the particle updates its velocity with the following equation

$$\begin{aligned} \boldsymbol{v_i}(t+1) = \boldsymbol{v_i}(t) &+ c_1\phi_1\big(\boldsymbol{pb_i}(t) - \boldsymbol{x_i}(t)\big) \\ &+ c_2\phi_2\big(\boldsymbol{gb}(t) - \boldsymbol{x_i}(t)\big) \end{aligned} \tag{5}$$

and positions with the following equation

$$\boldsymbol{x_i}(t+1) = \boldsymbol{x_i}(t) + \boldsymbol{v_i}(t+1) \tag{6}$$

where $\boldsymbol{v_i}$ is the particle velocity, $\boldsymbol{x_i}$ is the current particle (solution) and $\boldsymbol{Pb_i}$ and $\boldsymbol{gb}$ are defined as stated before. ϕ_1 and ϕ_2 are the random numbers between (0, 1). C_1 and C_2 are the learning factors, usually $c1 = c2 = 2$. The details of parameter optimisation of FNT using PSO algorithm is depicted in Algorithm 2 (see Fig. 4).

When using PSO algorithm, a possible solution to optimisation problem is represented by the position of a

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Algorithm 2:

Data: Weights w_i and activation function parameters
Result: Best solution **pg**
foreach *particle i in S* **do**
  Initialise all particle's position and velocity randomly:
  $\mathbf{x_i} \sim U(x_{min}, x_{max})$,
  $\mathbf{v_i} \sim Random(-v_{max}/3, v_{max}/3)$;
  Initialise particle's best position:
  $\mathbf{pb_i} = \mathbf{x_i}$;
  Update the global best position:
  if $f(\mathbf{pb_i}) < f(\mathbf{gb})$ **then**
    $\mathbf{gb} = \mathbf{pb_i}$;
  end
end
repeat
  foreach *particle i in S* **do**
    if $f(\mathbf{x_i}) < f(\mathbf{pb_i})$ **then**
      $\mathbf{pb_i} = \mathbf{x_i}$;
    end
    if $f(\mathbf{pb_i}) < f(\mathbf{gb})$ **then**
      $\mathbf{gb} = \mathbf{pb_i}$;
    end
  end
  foreach *particle i in S* **do**
    Pick random numbers:$\phi_1, \phi_2 \sim U(0, 1)$;
    Update the particle's velocity as Equation.5 ;
    Update the particle's position as Equation.6 ;
  end
  $it = it + 1$;
until $it > MAXITERATIONS$;

Fig. 4 *Parameter optimisation using PSO algorithm*

particle. The parameters encoded in the best FNT tree, the weights and flexible activation function parameters, are formulated as a particle. With fixed population s, all particles is initialised randomly. The particle's position is initialised with random vector $\boldsymbol{x_i} \sim U(x_{\min}, x_{\max})$, where $x_{\min}$ and $x_{\max}$ are the lower and upper boundaries of the search space. Particle's velocity is clamped to a maximum velocity $V_{\max}$. PSO is an iterative process. At each step it, the fitness value of each particle is calculated firstly. If the fitness value is better than the best fitness value $\boldsymbol{pb}$ in history, $\boldsymbol{pb_i}$ will be replaced with current value $\boldsymbol{x_i}$. $\boldsymbol{gb}$ is also be updated with the best fitness value of all the particles. Then, the particle's velocity is calculated according to (5) and its position is updated according to(6). This processing will be repeated until the maximum iterations or minimum error criteria is attained.

3.2.3 General learning algorithm: With predefined F set and T set, a FNT based skin colour model M can be created and evolved via a general learning algorithm, which is combined the structure optimisation and parameter optimisation with details described in Algorithm 3.

3.3 Selection of important components

Whether there is a best colour space or which channels have greater contributions than others are still a problem for skin detection. There are numerous colour spaces, such as RGB, YCbCr, HSV and CIE-Lab etc. Although there exists transformations between colour spaces [40], many works have given comparative studies on performance of different colour space choice. At present, colour space and colour components are usually predefined for pixel-wise skin colour detection methods, which limits the generalisation and performance of skin detection methods. For FNT-based skin colour model, the nature of model construction procedure allows the model to identify important input features that is computationally efficient and effective. The mechanisms of feature selection in the model constructing procedure is illustrated in Algorithm 2 (see Fig. 4).

1. The components c_i in feature x_i are selected to formulate the FNT-based skin model with same probabilities.
2. The components which have more contributions to the objective function will be enhanced and have high opportunity to survive in the next generation by a evolutionary procedure.
3. The evolutionary operators, that is, crossover and mutation, provide a input selection method by which the FNT should select appropriate variables automatically.
4. Return selected features.

Algorithm 3: General learning of FNT-based skin colour model

Require: Function set F and features f (as T set)
Ensure: Optimal model M

1. Initialisation. Create an initial population randomly (FNT trees and the corresponding parameters), set initial values of parameters used in GP and PSO algorithms;
2. Structure optimisation. Optimal structure will be achieved by the natural tree variation operators as described in Section 3.1 with GP algorithm;
3. If a better structure is found, then go to step (4), otherwise go to step (2);
4. Parameter optimisation using PSO algorithm. The architecture of FNT is fixed in this step, and it is the best tree developed during the end of run of the structure search. The parameters (weights and flexible activation function parameters) encoded in the best tree formulate a particle;
5. If the maximum number of the local search is reached, or no better parameter vector is found for a significantly long time then go to step (6); otherwise go to step (4);
6. If satisfactory solution is found, then the algorithm is stopped; otherwise go to step (2);
7. **return** Optimal FNT-based skin model.

Table 2 Skin data for training and testing

Datasets	Skin images	Non-skin images	Skin pixels	Non-skin pixels
Compaq training	2000	2000	27 102 292	83 977 080
Compaq testing	2000	2000	23 951 422	87 641 050
ECU training	2000	2000	102 679 306	453 023 571
ECU testing	2000	2000	106 706 759	448 010 613

Table 3 Different models of FNT-based skin detection

Models	*F*-set	*T*-set	Feature-set
*M*1	$\{+_2, +_3\}$	$\{x_1, x_2, x_3\}$	f3-RGB
*M*2	$\{+_2, +_3\}$	$\{x_1, x_2, x_3\}$	f3-CIE
*M*3	$\{+_2, +_3\}$	$\{x_1, x_2, \ldots, x_6\}$	f6-RC
*M*4	$\{+_2, +_3\}$	$\{x_1, x_2, \ldots, x_6\}$	f6-RY
*M*5	$\{+_2, +_3\}$	$\{x_1, x_2, \ldots, x_6\}$	f6-YC
*M*6	$\{+_2, +_3\}$	$\{x_1, x_2, \ldots, x_9\}$	f9-RYH
*M*7	$\{+_2, +_3\}$	$\{x_1, x_2, \ldots, x_9\}$	f9-RYC
*M*8	$\{+_2, +_3\}$	$\{x_1, x_2, \ldots, x_{12}\}$	f12-ALL
*M*9	$\{+_2, +_3, \ldots +_6\}$	$\{x_1, x_2, \ldots, x_{12}\}$	f12-ALL

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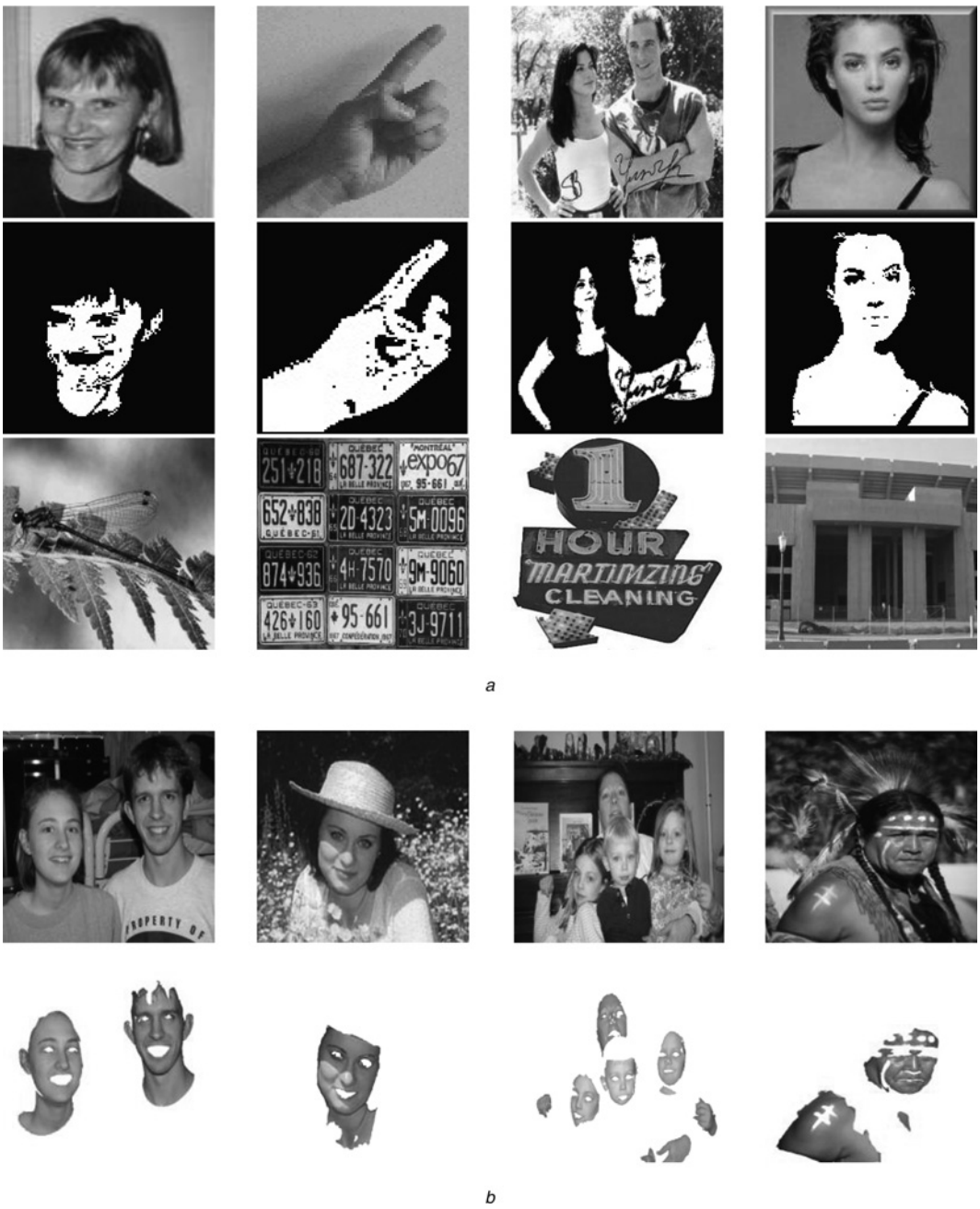

Fig. 5 *Example images in dataset*

Skin images with its' ground truth and non-skin images of Compaq dataset are shown in (a), and (b) shows colour images with corresponding skin segmented regions in ECU dataset
a Sample images in Compaq set
b Sample images in ECU set

For example, given input features comprised by nine channels of RGB, HSV and YCbCr colour spaces, the optimised model will select R, G, B, Y, Cr and S components as the important variables (see Fig. 7*a*).

4 Experimental results

In the experiment, the FNT-based skin colour model is built and evaluated with state of the art methods on the commonly used dataset.

Table 4 Comparison of FNT-based skin colour models

	*M*1, %	*M*2, %	*M*3, %	*M*4, %	*M*5, %	*M*6, %	*M*7, %	*M*8, %	*M*9, %
TPR	93.56	96.59	95.20	94.39	93.46	91.59	95.11	86.70	88.97
FPR	6.31	70.90	6.82	6.56	5.24	4.93	6.84	10.53	3.87
Acc	93.62	62.85	94.19	93.91	94.11	93.33	94.14	88.08	92.55
F-score	93.62	72.22	94.25	93.95	94.08	93.21	94.19	87.92	92.28

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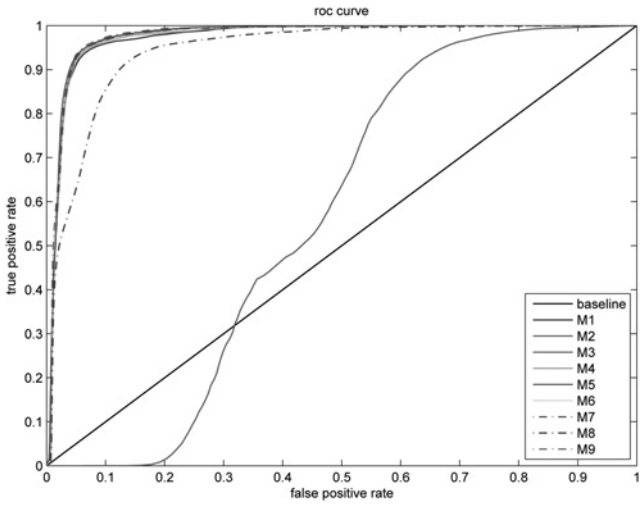

Fig. 6 *ROC curves of nine FNT skin models*

4.1 Datasets

The Compaq [32] and ECU [9] datasets are used for training and testing of FNT-based skin detection. There are 4670 skin images with ground-truth data and 4997 non-skin images in Compaq set, 4000 skin segmented images in ECU set. Both of these two sets consist of original colour images which are diversity in backgrounds, illumination and skin types. We randomly selected images from each dataset for training and testing. The detailed information of datasets are summarised in Table 2 and some example images in Compaq and ECU are shown in Fig. 5.

Four most popular colour spaces, that is, RGB, YCbCr, HSV and CIE-Lab, are used in experiments. Feature vectors extracted from each pixel of each colour space can be created by combined all the components of colour spaces. That is $x = (R, G, B, Y, Cb, Cr, H, S, V, L, a, b)$ where R, G, B are channels of RGB space, so does other colour spaces. Thus, the amount of data for the training and testing is over 17 GB in Compaq and ECU datasets.

For training, skin pixels are taken from the skin regions of segmented images in Compaq and ECU datasets and non-skin pixels from non-skin images (mask images) in Compaq set and the complement part of the skin segmented images in ECU set, respectively. For testing, a set of float values is first obtained via FNT-skin model on pixels. Then, a threshold value is applied on the float values (the value is 0.5 in experiment). If the FNT value of a pixel is greater than the threshold, the pixel is a skin pixel, otherwise is non-skin pixel. To evaluate the performance of our method and others, the skin segmented images obtained via skin classifiers are compared pixel-wise with the corresponding ground-truth images. No extra preprocessing and postprocessing steps are used. The performance of skin detection methods are measured in terms of TPR, FPR, Acc and F-score.

4.2 Different FNT-based skin colour models

Given predefined function set F and terminal instruction set T, a specific FNT-based skin model will be trained and constructed using skin and non-skin training data. In the experiment, we first construct nine FNT-based skin models with different F and T set. Details of these nine models are

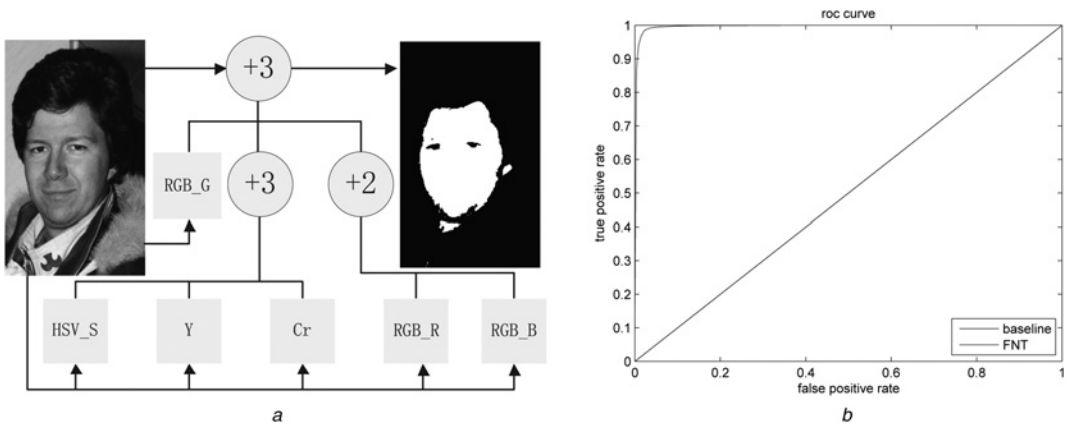

Fig. 7 *FNT-based skin colour detection*

The model allows over-layer connections, can automatically select the input features, give higher accuracy and lower FPR in pixel-wise skin detection
a Framework of FNT-based skin detection
b ROC curve of skin detection

Table 5 Comparison of skin colour classifiers

	HB, %	GMM, %	EB, %	MLP, %	Proposed, %
TPR[a]	91.47	89.12	84.68	90.12	90.51
FPR[a]	10.19	10.80	19.52	12.24	6.70
Acc[a]	90.17	89.18	81.38	89.57	91.82
TPR[b]	83.27	80.83	90.43	88.91	87.14
FPR[b]	10.29	15.75	24.05	13.70	9.96
Acc[b]	88.47	83.59	78.73	87.63	90.53

[a]For Compaq data set.
[b]For ECU data set.

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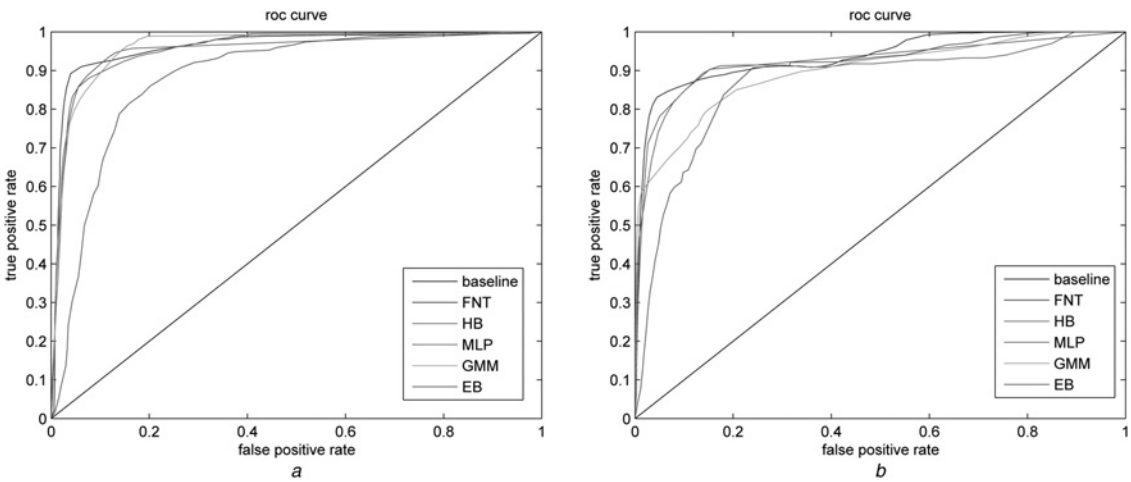

Fig. 8 *ROC curves of experimental results*

a ROC curve on Compaq dataset
b ROC curve on ECU dataset

illustrated in Table 3. Eight types of *T* set are constructed as follows:

- *f3-RGB:* combine R, G and B channels of RGB colour space;
- *f3-CIE:* combine L, a, and b channels of CIE-Lab colour space;
- *f6-RC:* add L, a,and b channels to f3-RGB;
- *f6-RY:* add Y, Cb, and Cr channels to f3-RGB;
- *f6-YC:* add Y, Cb, and Cr channels f3-CIE;
- *f9-RYH:* add H, S, V of HSV colour space to f6-RY;
- *f9-RYC:* add L, a, b channels to f6-RY;
- *f12-ALL:* All channels of RGB, YCbCr, HSV and CIE-Lab;

Fig. 9 *Sample results of skin classifiers*

a and *b* Original image and its ground-truth is shown in column
The detection results of
c Our method
d HB
e GMM
f MLP
g EB are depicted from left to right column

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With two types of F set, nine FNT skin models are constructed on eight T set. Table 3 shows the nine models. The F set $\{+_2, +_3\}$ is assigned to $M1$ through $M9$ models. The F set of $M9$ models is more complicated, which has the form of $\{+_2, +_3, \ldots, +_6\}$. Different combinations of F and T sets can help us to find the more reliable FNT skin models for skin colour detection.

With a subset data (30% of Compaq data), these nine models in Table 3 are trained and tested. The optimal structure of $M1$ model is similar to Fig. 2*a* and Fig. 7*a* shows the optimal structure of $M6$ model based on total training data. Table 4 shows the performance in terms of TPR, FPR, Acc and F-score. As shown in Table 4, most of FNT-based skin models achieve good performance in skin pixel classification. For example, $M1$ model achieves Acc of 93.62% and FPR of 6.31%. $M6$ model obtains the most lower FPR (4.93%) while $M3$ model obtains a higher TPR of 95.20%. The performance of $M2$ and $M8$ model is

Fig. 10 *Original colour images and skin segmentation results*

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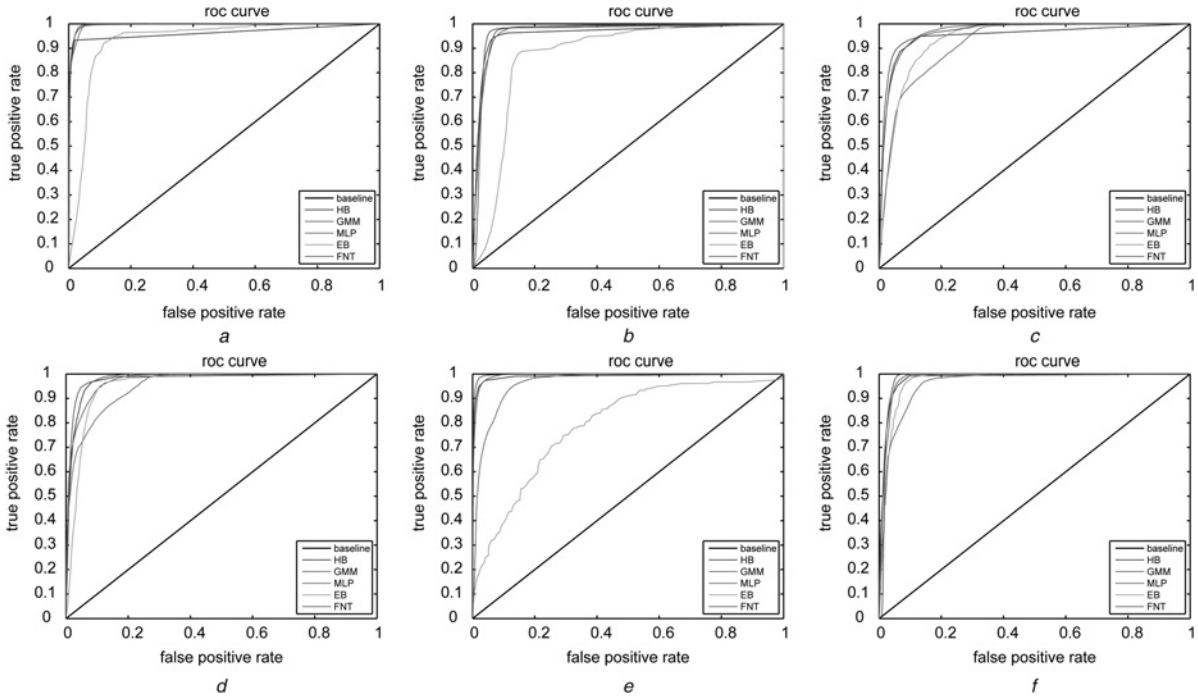

Fig. 11 *ROC curves of skin classifiers on sample images*

a–f The corresponding ROC curves of skin detection results on sample images depicted from first row to last row in Fig. 9

disappointed. If only CIE-Lab was used, *M*2 model gives highest TPR (96.59%) with a very high FPR (70.90%). Although all channels of four colour spaces are used, *M*8 gives poor detection result.

Fig. 6 depicts the ROC curve of nine FNT skin models. The ROC curve shows that *M*2 model and *M*8 model give poor performance compared with the reminder models, which is in accordance with the results in Table 4. Among these nine models, *M*6 model achieves better performance for trade-off between accuracy and FPR, which will be used in the comparison with other skin detection methods. Fig. 7*a* illustrates the framework of *M*6 model and Fig. 7*b* shows the ROC curve of the face detection result.

4.3 Comparison with other methods

Histogram-based Bayes classifier (HB), Gaussian mixture model (GMM), elliptic boundary model (EB) and MLP are commonly used in pixel-wised skin detection, which have sate of the art performance in reported work. In experiment, the parameters of these methods are chosen as follows:

- HB is constructed using our training data with histogram size of 256 bins per channel in RGB colour space.
- GMM has 16 Gaussians in skin mixture model, so does the non-skin class. Both of models use parameters published by Jones and Rehg [32].
- Elliptic boundary model (EB) uses parameters of Lee and Yoo [34].
- MLP is trained using our training data. we trained the network using Levenberg–Marquardt algorithm and back propagation algorithm. Different network size were investigated, and the best result is reserved.
- Proposed method with *M*6 model is trained with predefined parameters listed as follows: population size is 50, crossover probability and mutation probability is 0.4 and 0.03, both of c_1 and c_2 in PSO algorithm is 2.0.

As shown in Table 5, our method achieves an accuracy of 91.82% and FPR of 6.70% on Compaq dataset, which is better than other methods. Although HB method has a higher TPR (91.47%) than proposed method (90.51% of TPR), its accuracy and FPR are not as good as ours. Fig. 8 shows the ROC curves of experimental results on Compaq and ECU datasets. According to the left part of Fig. 8*a*, the FNT-based skin model achieves better performance in terms of accuracy and FPR, but has lower TPR than HB method. MLP and GMM methods obtains similar performance in Fig. 8*a*, which are better than EB method. Above analysis is in accordance with illustration in Table 5. EB method achieves the worst results with an Acc of 81.38% and FPR of 19.52% among skin colour classifiers on Compaq dataset. Fig. 8*b* depicts the ROC curve of skin detection results on ECU datset, which is also in agreement with the results in Table 5.

Fig. 9 lists sample results of skin colour detection, and Fig. 10 shows supplementary skin detection results on Compaq and ECU datasets. The corresponding ROC curves of skin detection on sample images of Fig. 9 are shown in Fig. 11. The first column (a) in Fig. 9 shows the original colour images. The ground-truth image and results of skin colour classifiers are illustrated in column (b) through (g). According to the ground-truth image in Fig. 9, our method classifies the skin pixel and non-skin pixel more correctly with lower false positive detection than other methods, which agrees with the ROC curve analysis. For example, the conclusion that HB, GMM and our method achieve better performance than MLP and EB methods could be decided according to ROC analysis of Fig. 11*b*, which is in accordance with the visual perception in Fig. 9.

5 Conclusions

In this paper, a novel skin colour model based on the FNT is introduced for pixel-wise skin colour detection. The

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contributions are: (i) without considering the selection of suitable colour spaces, the proposed method selects important components of colour spaces automatically. Given features comprised of all channels of commonly used colour spaces, the optimised skin model retains the most important ones, which present a solution of the choice of colour spaces and (ii) FNT-based skin detection method achieves higher accuracy and lower FPRs compared with state of the art methods. In the training stage, the individuals which have simple structures and similar model accuracy will be selected, which makes the structure of our model is generally simpler. The optimised hybrid tree structure guarantees state of the art performance in skin pixel classification. The training of the model is time consuming for the optimisation of structure and parameters, but it is not a problem with high-performance computers and parallel algorithm.

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